

Gold Future/ETF Analysis

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Table 1: New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.98218	-9.0480×10^{-5}	0.92459	0.00269
2-Month	0.98709	-1.2547×10^{-4}	0.92244	0.00275
6-Month	0.98870	-9.0354×10^{-5}	0.91718	0.00285
12-Month	0.92794	9.3489×10^{-5}	0.76020	0.00520

Slopes are very close to 1, apart from the 12-month contract. This may be a result of the lower liquidity of the longer contract and the smaller impact of daily gold price movements on longer-dated futures.

Table 2: Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.94314	2.4191×10^{-5}	0.80947	0.00530
2-Month	0.94301	1.0000×10^{-5}	0.80911	0.00530
6-Month	0.94348	3.1797×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.2333×10^{-5}	0.80984	0.00529

The slopes are also very close to 1, but consistently lower than the newer data. The R^2 values are also significantly lower than the newer data. This may be due to the smaller sample size of the older data and lower liquidity.

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Table 3: New Data: Slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.96061	0.96608	0.96757	0.91176
	5	0.99723	1.00073	0.99971	0.94958
	10	0.99556	0.99782	0.99821	0.97219
	15	0.99821	1.00003	0.99906	0.98087
R^2	1	0.87699	0.87470	0.87272	0.73603
	5	0.96413	0.96345	0.96008	0.87631
	10	0.98020	0.97931	0.97586	0.93513
	15	0.98603	0.98532	0.98191	0.95316

The slopes and R^2 values generally increase with longer holding periods but decrease with longer-dated contracts.

Table 4: Leung & Ward (2015) Data: Slopes and R^2 from regressions of futures returns vs. gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of \$1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one. The GLD RMSE is \$6.00 for the out of sample period.

Table 5: New Data: Weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

Portfolio	w_0	w_1	w_2	w_6	w_{12}	RMSE _{in}	RMSE _{out}
1-m	0.00285	0.99711				\$3.27	\$9.18
2-m	0.00723		0.99273			\$4.22	\$10.64
12-m	0.04846				0.95208	\$19.36	\$24.98
1-m, 2-m	0.00189	1.22277	-0.22470			\$3.23	\$8.91
1-m, 6-m	0.00198	1.04254		-0.04456		\$3.24	\$8.90
1-m, 12-m	0.00234	1.01049			-0.01286	\$3.26	\$9.16
2-m, 6-m	0.00367		1.22476	-0.22854		\$3.81	\$9.41
2-m, 12-m	0.00390		1.08780		-0.09172	\$3.87	\$10.60
6-m, 12-m	0.00919			1.66574	-0.67435	\$7.00	\$23.51
1-m, 2-m, 6-m	0.00184	1.20060	-0.19291	-0.00956		\$3.23	\$8.88
1-m, 2-m, 12-m	0.00204	1.28345	-0.29678		0.01126	\$3.22	\$8.84
1-m, 6-m, 12-m	0.00210	1.10380		-0.17056	0.06457	\$3.21	\$8.31
2-m, 6-m, 12-m	0.00369		1.22868	-0.23595	0.00348	\$3.81	\$9.37
1-m, 2-m, 6-m, 12-m	0.00196	1.24924	-0.17947	-0.13469	0.06287	\$3.20	\$8.30

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Table 6: Leung & Ward (2015) Data: Weights and in/out of sample RMSEs for portfolios of 1 and 2 futures contracts.

Portfolio	w_0	w_{short}	w_{long}	RMSE _{in}	RMSE _{out}
1-m	-0.01071	1.01071		\$6.63	\$3.34
2-m	-0.04835	1.04835		\$12.22	\$6.12
12-m	-0.06842	1.06842		\$15.11	\$5.71
1-m, 2-m	-0.00088	1.27315	-0.27227	\$6.27	\$2.79
1-m, 6-m	-0.00171	1.23899	-0.23729	\$6.27	\$2.98
1-m, 12-m	-0.00021	1.19336	-0.19315	\$6.29	\$3.00
2-m, 6-m	-0.04079	5.06602	-4.02523	\$10.27	\$9.37

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Table 7: New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.00061	-8.8911×10^{-6}	0.25954	0.79523	0.97140	0.00185
UGL	2	2.02088	-1.9483×10^{-4}	2.77734	0.00552	0.96438	0.00349
GLL	-2	-2.01429	1.8890×10^{-4}	-1.92313	0.05457	0.96497	0.00345
DGLD	-2	-1.97565	-8.4896×10^{-5}	0.46841	0.63954	0.40656	0.02098
UGLDF	3	3.00632	4.5874×10^{-4}	0.05822	0.95358	0.25518	0.04463

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Table 8: Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1.00540	-1.6427×10^{-6}	1.42692	0.15382	0.98060	0.00163
UGL	2.00572	-1.3107×10^{-4}	0.73319	0.46357	0.97930	0.00337
GLL	-2.00556	-1.0550×10^{-4}	0.67089	0.50240	0.97673	0.00358
UGLD	2.99358	-1.9860×10^{-4}	0.45211	0.65133	0.98484	0.00421
DGLD	-2.97528	-1.6912×10^{-5}	0.97442	0.33019	0.95256	0.00752

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The t-statistics far from zero for UGL and GLL indicate that the slopes are statistically significantly different from the specified leverage ratios.

Table 9: New Data: Weights and in/out of sample RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	Portfolio	w_0	w_1	w_2	w_6	w_{12}	RMSE _{in}	RMSE _{out}
2x Leveraged	1-m	-1.62098	2.62436				\$58.87	\$331.76
	2-m	-1.61086		2.61411			\$56.14	\$336.05
	6-m	-1.56497			2.56792		\$45.85	\$357.43
	12-m	-1.49557				2.49767	\$46.46	\$375.93
	1-m, 2-m	-1.56365	-10.84072	13.40753			\$50.70	\$353.93
	1-m, 6-m	-1.54357	-1.39403		3.94189		\$45.17	\$365.64
3x Leveraged	1-m	-4.03105	5.03349				\$188.10	\$1036.37
	2-m	-4.01308		5.01515			\$183.04	\$1042.61
	6-m	-3.92928			4.92945		\$161.06	\$1076.52
	12-m	-3.79391				4.79256	\$147.85	\$1116.16
	1-m, 2-m	-3.85870	-35.37062	40.23119			\$165.26	\$1085.79
	1-m, 6-m	-3.79649	-7.19724		11.99866		\$149.02	\$1111.73

Table 10: Leung & Ward (2015) Data: RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	Portfolio	RMSE _{in}	RMSE _{out}
+2x (UGL)	1-m	\$153.20	\$41.49
	2-m	\$140.41	\$49.57
	6-m	\$138.63	\$47.46
	12-m	\$134.51	\$48.30
+3x (UGLD)	1-m	\$555.49	\$111.51
	2-m	\$529.91	\$125.99
	6-m	\$526.58	\$122.33
	12-m	\$518.97	\$123.83